

Algebra II with Trigonometry

Chapter 11.3

Mental Math (GPT-5.4)

Saved tutoring response with MathJax rendering. Original PDF file:
[4666324_alg-2trig-h-chp-11-3-note-docx.pdf](#)

- 1. Definition Check** A hyperbola is defined using the sum or the difference of distances from two fixed points?
- 2. Orientation from Equation** For $\frac{x^2}{16} - \frac{y^2}{9} = 1$, does the hyperbola open left-right or up-down?
- 3. Vertices Quick Find** For $\frac{x^2}{25} - \frac{y^2}{4} = 1$, what are the vertices?
- 4. No Intercepts?** For $\frac{x^2}{9} - \frac{y^2}{16} = 1$, is there a y-intercept?
- 5. Transverse Axis Length** For $\frac{y^2}{36} - \frac{x^2}{4} = 1$, what is the length of the transverse axis?
- 6. Asymptotes from Standard Form** For $\frac{x^2}{4} - \frac{y^2}{25} = 1$, what are the asymptotes?
- 7. Vertical Hyperbola** For $\frac{y^2}{49} - \frac{x^2}{9} = 1$, what are the vertices?
- 8. Find the Foci** For $\frac{x^2}{9} - \frac{y^2}{16} = 1$, what are the foci?
- 9. Equation Type Recognition** Does the equation $9x^2 - 16y^2 = 1$ represent a hyperbola, ellipse, or parabola?
- 10. Build the Equation** A hyperbola has vertices $(\pm 2, 0)$ and foci $(\pm 6, 0)$. What is its standard equation?